CoAP pour l'IoT — TP complet avec serveur Python

MODULE 1

1.1 Lire l'heure — GET

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/time
2026-09-29T09:03:28Z
```

1.2 Lire la température (ressource simulée)

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/temp
23.2
```

1.3 Imposer une température avec PUT

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client -vv -m put --payload "19.0" coap://127.0.0.1/temp
DEBUG:coap:Sending request - Token: 7e59, Remote: <_Connection at 0x105cfbc70 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsiz
e=0>>, active>
DEBUG:coap:Sending message <aiocoap.Message at 0x105cfbfd0: CON PUT (MID 56625, token 7e59) remote <_Connection at 0x105cfbc70 on transport <_SelectorDatagr
amTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 option(s), 4 byte(s) payload>
DEBUG:coap:Exchange added, message ID: 56625.
DEBUG:coap:Incoming message <aiocoap.Message at 0x105c95130: ACK 2.04 Changed (MID 56625, token 7e59) remote <_Connection at 0x105cfbc70 on transport <_Sele
ctorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>>
DEBUG:coap:Exchange removed, message ID: 56625.
DEBUG:coap:Received Response: <aiocoap.Message at 0x105c95130: ACK 2.04 Changed (MID 56625, token 7e59) remote <_Connection at 0x105cfbc70 on transport <_Se
lectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>>
DEBUG:coap:Response <aiocoap.Message at 0x105c95130: ACK 2.04 Changed (MID 56625, token 7e59) remote <_Connection at 0x105cfbc70 on transport <_SelectorData
gramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>> matched to request <Pipe at 0x105cfbbb0 around <aiocoap.Message at 0x105cfbfd0: CON PUT (
MID 56625, token 7e59) remote <_Connection at 0x105cfbc70 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 op
tion(s), 4 byte(s) payload> with 2 callbacks>
DEBUG:coap:Shutting down context
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tcp.TCPClient object at 0x105c95340>
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tls.TLSClient object at 0x105c953d0>
```

1.4 Allumer / éteindre la LED

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/led
off
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client -m put -vv --payload "on" coap://127.0.0.1/led
DEBUG:coap:Sending request - Token: 2b39, Remote: <_Connection at 0x107fa3ca0 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsiz
e=0>>, active>
DEBUG:coap:Sending message <aiocoap.Message at 0x107fa3fd0: CON PUT (MID 51431, token 2b39) remote <_Connection at 0x107fa3ca0 on transport <_SelectorDatagr
amTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 option(s), 2 byte(s) payload>
DEBUG:coap:Exchange added, message ID: 51431.
DEBUG:coap:Incoming message <aiocoap.Message at 0x107f3d130: ACK 2.04 Changed (MID 51431, token 2b39) remote <_Connection at 0x107fa3ca0 on transport <_Sele
ctorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>>
DEBUG:coap:Exchange removed, message ID: 51431.
DEBUG:coap:Received Response: <aiocoap.Message at 0x107f3d130: ACK 2.04 Changed (MID 51431, token 2b39) remote <_Connection at 0x107fa3ca0 on transport <_Se
lectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>>
DEBUG:coap:Response <aiocoap.Message at 0x107f3d130: ACK 2.04 Changed (MID 51431, token 2b39) remote <_Connection at 0x107fa3ca0 on transport <_SelectorData
gramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>> matched to request <Pipe at 0x107fa3d60 around <aiocoap.Message at 0x107fa3fd0: CON PUT (
MID 51431, token 2b39) remote <_Connection at 0x107fa3ca0 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 op
tion(s), 2 byte(s) payload> with 2 callbacks>
DEBUG:coap:Shutting down context
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tcp.TCPClient object at 0x107f3d340>
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tls.TLSClient object at 0x107f3d3d0>
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client -m put --payload "peut-etre" coap://127.0.0.1/led
4.00 Bad Request
use 'on' ou 'off'
```

1.5 Envoyer une alerte avec POST

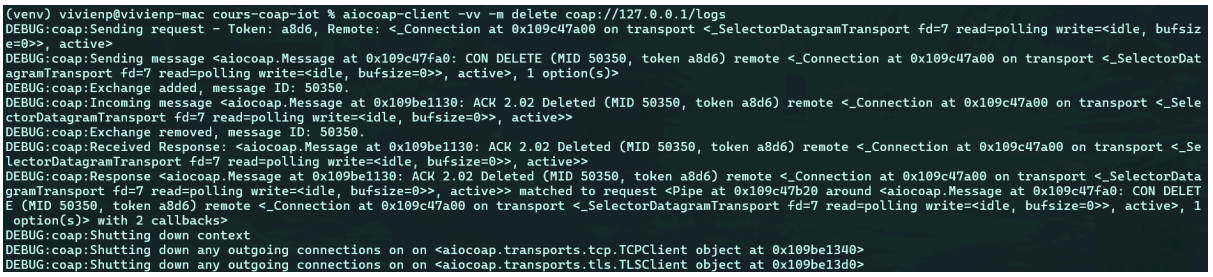
```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client -m post -vv --payload "ALERTE: Temperature elevee 35C" coap://127.0.0.1/logs
DEBUG:coap:Sending request - Token: 48e5, Remote: <_Connection at 0x1084aba00 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsiz
e=0>>, active>
DEBUG:coap:Sending message <aiocoap.Message at 0x1084abfd0: CON POST (MID 14485, token 48e5) remote <_Connection at 0x1084aba00 on transport <_SelectorDatag
ramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 option(s), 30 byte(s) payload>
DEBUG:coap:Exchange added, message ID: 14485.
DEBUG:coap:Incoming message <aiocoap.Message at 0x108445130: ACK 2.01 Created (MID 14485, token 48e5) remote <_Connection at 0x1084aba00 on transport <_Sele
ctorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 option(s)>
DEBUG:coap:Exchange removed, message ID: 14485.
DEBUG:coap:Received Response: <aiocoap.Message at 0x108445130: ACK 2.01 Created (MID 14485, token 48e5) remote <_Connection at 0x1084aba00 on transport <_Se
lectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 option(s)>
DEBUG:coap:Response <aiocoap.Message at 0x108445130: ACK 2.01 Created (MID 14485, token 48e5) remote <_Connection at 0x1084aba00 on transport <_SelectorData
gramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 option(s)> matched to request <Pipe at 0x1084abb20 around <aiocoap.Message at 0x1084abf
d0: CON POST (MID 14485, token 48e5) remote <_Connection at 0x1084aba00 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>,
 active>, 1 option(s), 30 byte(s) payload> with 2 callbacks>
Location options indicate new resource: /logs/1
DEBUG:coap:Shutting down context
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tcp.TCPClient object at 0x108445340>
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tls.TLSClient object at 0x1084453d0>
```

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client -m post -v --payload "ALERTE: Temperature elevee 35C" coap://127.0.0.1/logs
Location options indicate new resource: /logs/2
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/logs
1: [11:15:11] ALERTE: Temperature elevee 35C
2: [11:15:26] ALERTE: Temperature elevee 35C
```

1.6 Ressource inexistante — erreur 4.04

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/inexistant
4.04 Not Found
```

1.7 DELETE — à vous de coder !



```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client -vv -m delete coap://127.0.0.1/logs
DEBUG:coap:Sending request - Token: a8d6, Remote: <_Connection at 0x109c47a00 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsiz
e=0>>, active>
DEBUG:coap:Sending message <aiocoap.Message at 0x109c47fa0: CON DELETE (MID 50350, token a8d6) remote <_Connection at 0x109c47a00 on transport <_SelectorDat
agramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1 option(s)>
DEBUG:coap:Exchange added, message ID: 50350.
DEBUG:coap:Incoming message <aiocoap.Message at 0x109be1130: ACK 2.02 Deleted (MID 50350, token a8d6) remote <_Connection at 0x109c47a00 on transport <_Sele
ctorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>>
DEBUG:coap:Exchange removed, message ID: 50350.
DEBUG:coap:Received Response: <aiocoap.Message at 0x109be1130: ACK 2.02 Deleted (MID 50350, token a8d6) remote <_Connection at 0x109c47a00 on transport <_Se
lectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>>
DEBUG:coap:Response <aiocoap.Message at 0x109be1130: ACK 2.02 Deleted (MID 50350, token a8d6) remote <_Connection at 0x109c47a00 on transport <_SelectorData
gramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>> matched to request <Pipe at 0x109c47b20 around <aiocoap.Message at 0x109c47fa0: CON DELET
E (MID 50350, token a8d6) remote <_Connection at 0x109c47a00 on transport <_SelectorDatagramTransport fd=7 read=polling write=<idle, bufsize=0>>, active>, 1
 option(s)> with 2 callbacks>
DEBUG:coap:Shutting down context
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tcp.TCPClient object at 0x109be1340>
DEBUG:coap:Shutting down any outgoing connections on on <aiocoap.transports.tls.TLSClient object at 0x109be13d0>
```

QUESTIONS / RÉSULTATS MODULE 1

Quel code retour obtenez-vous pour un GET sur /time ? 2.05

Après un PUT sur /temp avec la valeur 19.0, quel code retour ? 2.04

Que renvoie un PUT /led avec la valeur "peut-être" ? Pourquoi ? Un PUT /led avec "peut-être" renvoie 4.00 car le payload envoyé n'est pas l'état possible de la led qu'on a défini.

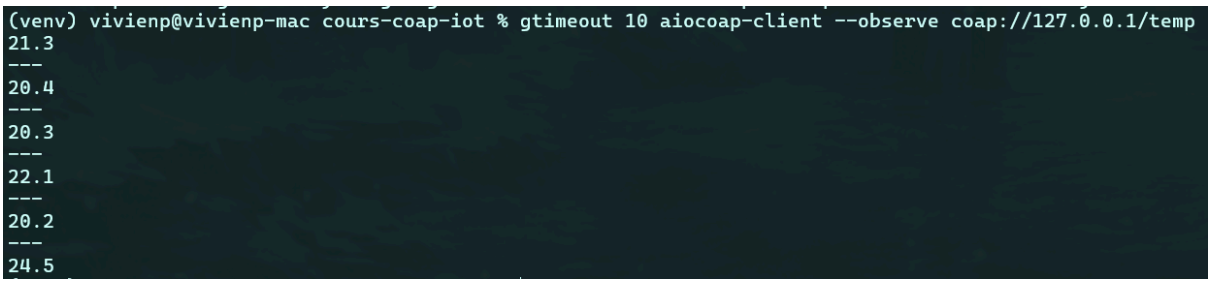
Quelle commande utiliseriez-vous pour éteindre la LED ? aiocoap-client -m put --payload "off" coap://127.0.0.1/led.

Après 2 POST successifs sur /logs, que renvoie le GET ? Il renvoie tous les logs précédents plus les deux logs récents.

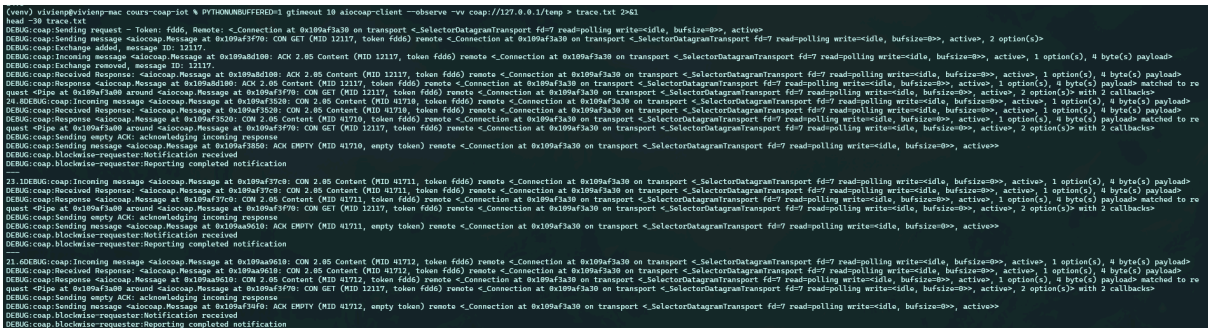
Quel code retour renvoie votre DELETE sur /logs ? 2.02

MODULE 2 : L'observation (la grande force de CoAP)

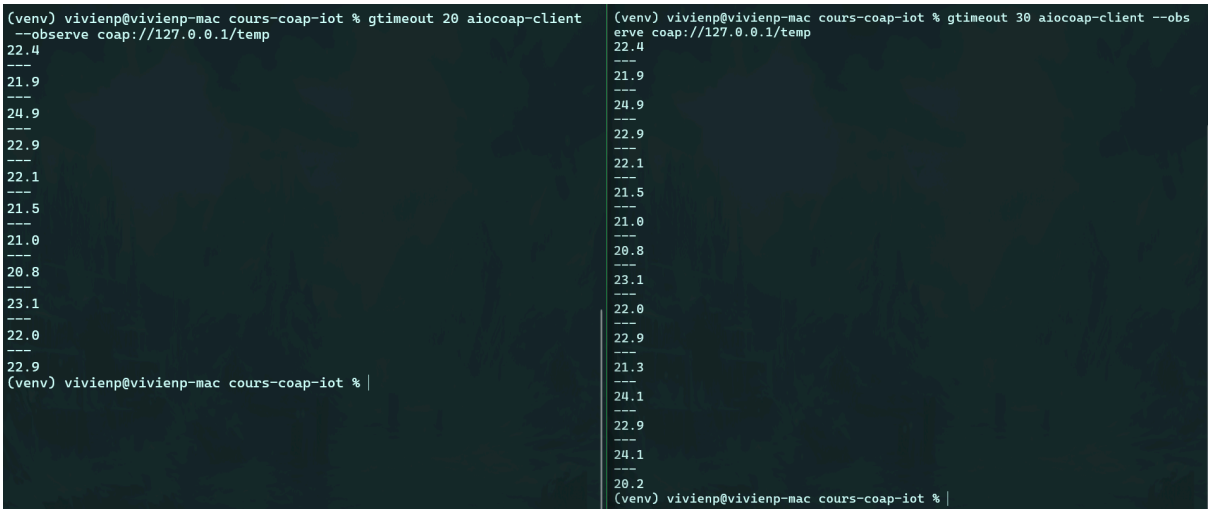
2.1 Démarrer une observation (10 s)



2.2 Voir les détails des messages avec --verbose



2.3 Test fondamental : plusieurs observateurs



2.4 Interrompre une observation

QUESTIONS / RÉSULTATS MODULE 2

Combien de notifications en 10 s (capteur qui change toutes les 2 s) ? 5

Deux clients observent /temp : reçoivent-ils tous les deux les notifications ? Oui

Polling 1 s vs observation (changement toutes les 5 s) sur 1 min : combien de messages dans chaque cas ? Polling envoie 120 messages, observation envoie 14 messages.

Donnez 2 usages de l'observation dans une serre connectée. Surveiller les capteurs (température, humidité,…) et surveiller les changements d'état des appareils avec tâche d'automatisation (LED, pompe,…).

Dans votre code serveur, quelle méthode déclenche l'envoi d'une notification ? self.updated_state()

MODULE 3 : Découverte des ressources et blockwise

3.1 La découverte : /.well-known/core

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/.well-known/core
</temp>;obs,</led>,</logs>,</time>,</.well-known/core>;ct="40",<https://christian.amsuess.com/tools/aiocoap/#version-0.4.7>;rel="impl-info"
```

3.2 Ajouter une arborescence de capteurs — exercice de code

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/sensors/room1/temperature
23.5
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/.well-known/core
</temp>;obs,</led>,</logs>,</time>,</sensors/room1/temperature>,</sensors/room1/humidity>,</sensors/room1/light>,</.well-known/core>;ct="40",<ht
tps://christian.amsuess.com/tools/aiocoap/#version-0.4.7>;rel="impl-info"
```

3.3 Le blockwise (découpage en blocs)

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://127.0.0.1/biglog | wc -c
    1784
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client -vv coap://127.0.0.1/biglog 2>&1 | grep -i block
DEBUG:coap.blockwise-requester:Response with Block2 option received, number = 1, more = 0, size_exp = 6.
```

3.5 Multicast CoAP

J'étais sur mon propre réseau, j'ai donc bind en mettant l'adresse IP de ma machine et appelé sur cette IP au lieu du localhost.

```
(venv) vivienp@vivienp-mac cours-coap-iot % aiocoap-client coap://192.168.1.74/.well-known/core
</temp>;obs,</led>,</logs>,</time>,</sensors/room1/temperature>,</sensors/room1/humidity>,</sensors/room1/light>,</biglog>,</.well-known/core>;c
t="40",<https://christian.amsuess.com/tools/aiocoap/#version-0.4.7>;rel="impl-info"
```

3.6 Passerelle CoAP ↔ HTTP (pour aller plus loin)

```
(venv) vivienp@vivienp-mac cours-coap-iot % python client.py
Temperature: 22.9
```

QUESTIONS / RÉSULTATS MODULE 3

Combien de ressources dans "/.well-known/core" après avoir ajouté /sensors/room1/… ? 8 entrées.

Que signifie l'attribut ;obs dans l'annuaire ? La ressource est observable.

Quelle est la taille du payload de /biglog (octets) ? En combien de blocs est-il téléchargé avec des blocs de 256 octets ? 1784 octets, soit 7 blocs de 256 octets.

Pourquoi le blockwise est-il essentiel pour une caméra IoT qui envoie des images ? Une image peut dépasser largement un diagramme UDP.

Quel est l'avantage du multicast pour installer 200 capteurs dans un bâtiment ? Une seule requête est nécessaire pour atteindre tous les capteurs.